Food Microbiology 91 (2020) 103530

Contents lists available at ScienceDirect

Food Microbiology

journal homepage: www.elsevier.com/locate/fm

Whole genome sequencing (WGS) fails to detect antimicrobial resistance (AMR) from heteroresistant subpopulation of *Salmonella enterica*

Ye Htut Zwe[a], Seow Fong Chin[b], Gurjeet Singh Kohli[b,c], Kyaw Thu Aung[d,e,f], Liang Yang[b,f,g], Hyun-Gyun Yuk[h,*]

[a] *Department of Food Science and Technology, National University of Singapore, Singapore*
[b] *Singapore Centre for Environmental Life Sciences Engineering, Nanyang Technological University, Singapore*
[c] *Alfred Wegener-Institut Helmholtz-Zentrum für Polarund Meeresforschung, Bremerhaven, Germany*
[d] *National Centre for Food Science, Singapore Food Agency, Singapore*
[e] *School of Chemical and Biomedical Engineering, Nanyang Technological University, Singapore*
[f] *School of Biological Sciences, Nanyang Technological University, Singapore*
[g] *School of Medicine, Southern University of Science and Technology, Shenzhen, China*
[h] *Department of Food Science and Technology, Korea National University of Transportation, Jeungpyeong-gun, Chungbuk, Republic of Korea*

ARTICLE INFO

Keywords:
Salmonella enterica
Whole genome sequencing
Heteroresistance
Antimicrobial resistance prediction
False negative errors

ABSTRACT

Due to rapidly falling costs, whole genome sequencing (WGS) is becoming an essential tool in the surveillance of antimicrobial resistance (AMR) in *Salmonella enterica*. Although there have been many recent works evaluating the accuracy of WGS in predicting AMR from a large number of *Salmonella* isolates, little attention has been devoted to deciphering the underlying causes of disagreement between the WGS genotype and experimentally determined AMR phenotype. This study analyzed the genomes of six *S. enterica* isolates previously obtained from raw chicken which exhibited disagreements between WGS genotype and AMR phenotype. A total of five WGS false negative predictions toward ampicillin, amoxicillin/clavulanate, colistin, and fosfomycin resistance were presented in conjunction with their corresponding empirical phenotypic and/or genetic evidence of heteroresistance. A further case study highlighting the inherent limitations of WGS to detect the underlying genetic mechanisms of colistin heteroresistance was presented. These findings implicate heteroresistance as an underlying cause for false negative WGS-based AMR predictions in *S. enterica* and suggest that widespread use of WGS in the surveillance of AMR in food isolates might severely underestimate true resistance rates.

1. Introduction

As the infections from antimicrobial-resistant organisms increasingly become a great public health concern (Uddin and Anh, 2018; Jo et al., 2017), rapid and accurate determination of the antimicrobial resistance (AMR) in bacterial isolates becomes imperative. Surveillance of antimicrobial-resistant microorganisms enables us to assess the nature, and extent of resistance in foodborne bacteria in food supply provides valuable information for risk assessment and guides regulatory decisions to preserve the effectiveness of valuable antimicrobials for humans and animals.

Traditional phenotypic antimicrobial susceptibility testing (AST) usually involves the *in vitro* measuring of the minimum inhibitory concentrations (MICs) using an array of serially diluted antimicrobials in broth or the diameter of inhibition zones around disks containing standard quantities of antimicrobials (Reller et al., 2009). Despite their simplicity, disadvantages include laborious procedures, the lack of harmonization that hinders data comparison between laboratories, practical limitations on the type of organism, and the number of agents that can be tested (Kahlmeter, 2014). An ideal characterization tool, therefore, should yield all of the necessary information on resistance in a rapid, comprehensive and universal manner, preferably, in a single step (Didelot et al., 2012). Theoretically, the genome of a pathogenic isolate should include virtually all the information needed to direct treatment and aid outbreak investigation measures.

Due to falling costs of the whole genome sequencing (WGS) technology, it is now possible to evaluate the bacterial genome in its entirety at affordable costs in a short time (Punina et al., 2015). This has facilitated the extensive use of WGS technology in applications such as outbreak detection and tracing (Leekitcharoenphon et al., 2014),

* Corresponding author. Department of Food Science and Technology, Korea National University of Transportation, 61 Daehak-ro, Jeungpyeong-gun, Chungbuk, 27909, Republic of Korea.
E-mail address: yukhg@ut.ac.kr (H.-G. Yuk).

https://doi.org/10.1016/j.fm.2020.103530
Received 30 April 2019; Received in revised form 28 March 2020; Accepted 22 April 2020
Available online 25 April 2020
0740-0020/

serotyping (Joensen et al., 2015; Zhang et al., 2015), and multilocus sequence typing (Larsen et al., 2012). Determining the full complement of genetic resistance determinants is an important application of the definitive genotypic data yielded from WGS. Moreover, it has the added advantages of discovering resistance to compounds not routinely tested and the ability to differentiate identical phenotypic resistance patterns caused by fundamentally different molecular mechanisms (McDermott et al., 2016). As a result, there has been a recent explosion in the literature of studies which demonstrated the use of WGS to predict AMR with promising concordance between genotype and phenotype in several pathogens (Gordon et al., 2014; McDermott et al., 2016; Neuert et al., 2018; Stoesser et al., 2013; Tyson et al., 2015). Although much praise has been lavished on WGS for accurately predicting phenotype from genotype in most cases, far less attention has been given to the few instances in which prediction errors occur. For WGS to be unequivocally accepted in the future as the gold-standard method of determining AMR, a firm grasp on the underlying reasons as to why disagreements between genotype from WGS data and experimentally determined phenotype may occur is required.

Heteroresistance is a phenomenon where a population of seemingly isogenic bacteria harbors a sub-population which exhibits a range of susceptibility to a particular antimicrobial and has been extensively reviewed elsewhere (El-Halfawy and Valvano, 2015). Although there has been recent literature on the topic of heteroresistance such as on its prevalence (Nicoloff et al., 2019) and underlying genetic mechanisms (Hjort et al., 2016; Nicoloff et al., 2019; Pournaras et al., 2010), to name a few, its negative effects on the accuracy of predicting AMR from WGS data has not been discussed. Thus, the objective of this study was to investigate the heteroresistance phenomenon causing false negative errors in WGS-based AMR predictions in six *S. enterica* strains previously isolated from raw chicken which exhibit heteroresistance toward ampicillin (Am), amoxicillin/clavulanate (Aug), colistin (Cl) and fosfomycin (Fos). A further case study exploring the molecular mechanism of Cl heteroresistance in *S. enterica* involving the *pmrD* gene copy number modulation and how it led to the failure of WGS to detect the resultant resistance was also provided.

2. Material and methods

2.1. Bacterial isolates and AST

Out of the 52 isolates of *S. enterica* previously isolated from raw chicken from supermarkets and wet markets of Singapore (Zwe et al., 2018), six particular strains comprising of two *S.* Saintpaul isolates, and one each of *S.* Brancaster, *S.* Gaminara, *S.* Albany and *S.* Stanley were chosen as subjects of this study due to their discrepancies in their phenotypic AMR patterns and their genotype. The strain ID, serotype and the MICs of the strains used in this study are summarized in Table 1.

Frozen stocks of the bacterial strains were stored at −80 °C in Cryoinstant vials with porous beads (DeltaLab, Barcelona, Spain). The cultures were activated by two consecutive 24 h transfers at 37 °C in 10 mL tryptone soya broth (TSB; Oxoid, Hampshire, England) and plated onto tryptone soya agar (TSA; Oxoid, Hampshire, England) to obtain discrete colonies to be suspended for AST. AST was performed by using the broth microdilution in dehydrated Neg MIC Panel Type 40 plates (MicroScan, Brea, CA, USA) according to the manufacturer's instructions. The MICs were interpreted as either resistant, intermediate or susceptible according to the EUCAST breakpoints (EUCAST, 2018) except tetracycline which was interpreted according to CLSI breakpoints (CLSI, 2017).

2.2. WGS of bacterial isolates and assembly

The DNA extraction was performed using the QIAamp DNA Mini Kit (Qiagen, Hilden, Germany) according to the manufacturer's instructions, sequenced using the Illumina HiSeq2500 platform (Illumina) and assembled using SPADES v3.5.1 (Bankevich et al., 2012) in Enterobase (https://enterobase.warwick.ac.uk/). The raw reads of the strains can be obtained from GenBank by these accession numbers: B27 (SAMN08848578), B28 (SAMN08848579), M24 (SAMN08848552), WD17 (SAMN08848566), WT5 (SAMN08848549), WW28B (SAMN08848596) under BioProject number PRJNA448475.

2.3. In silico prediction of AMR from WGS results

The assembled contig files were analyzed for the presence of genetic resistance determinants using the web-based ResFinder 3.2.0 (https://cge.cbs.dtu.dk/services/ResFinder/) (Zankari et al., 2012) and PointFinder 3.1.0 with database versions of 2020-02-27 and 2019-07-02, respectively. Resistance determinants were identified by setting the threshold for %ID at 90% and minimum length at 60% (default ResFinder 3.2.0 and PointFinder 3.1.0 values).

2.4. Comparing genotype to phenotype

Phenotypic susceptibility or resistance to a given antimicrobial was predicted *in silico* by the absence or presence, respectively, of a known corresponding resistance determinant from literature as compiled previously (Bush et al., 1995; Shaw et al., 1993; Stoesser et al., 2013). The phenotypically determined AMR profiles were then compared to the AMR profiles predicted *in silico* from the WGS data. In all instances of disagreement where genotype did not correlate with phenotype, both the phenotypic AST and ResFinder 3.2.0 analysis with WGS data were repeated at a lower threshold of %ID at 60% and minimum length at 40%. KmerResistance 2.2 analysis (Clausen et al., 2016, 2018) was performed further on isolates with false negative errors using the raw read files and default parameters (70% ID and 10% depth) to double-check that the loss of the 'missing' resistance gene did not occur due to the assembly process.

2.5. Determination of heteroresistance

Heteroresistance toward Aug in B27 and WT5, Am in WD17, Fos in WW28B were detected by the disk diffusion method as previously performed (Lo-Ten-Foe et al., 2007). The growth of colonies in the clear zone of inhibition was considered a sign of heteroresistance (El-Halfawy and Valvano, 2013).

Heteroresistance to Cl was performed using the population analysis profiling (PAP) assay on M24 (Cl-resistant *mcr-1* carrier), B27 (Cl heteroresistant-suspect), and B28 (Cl susceptible control) by turbidimetry method as previously described (El-Halfawy and Valvano, 2013) with slight modifications. Bacterial suspensions with a final bacterial load of approximately 5×10^5 CFU/mL were grown in TSB with increasing concentrations of Cl (Sigma, St. Louis, MO, USA) ranging from 0.125 to 32 μg/mL in 96-well plates at 37 °C for 20 h. OD_{595} increment at each concentration after the incubation was measured using the Thermo Scientific Multiskan FC Microplate Photometer (Thermo Fisher Scientific, Shanghai, China).

2.6. Determination of pmrD gene copy number

DNA was extracted from 24 h cultures of B27, B28, M24, and from M24, B27 grown in TSB in the presence of 2 μg/mL of Cl (1/2 MIC) using the GeneJET Genomic DNA Purification Kit (Thermo Fisher Scientific) following the manufacturer's instructions. The single copy chromosomal gene *hcaT* (Brandis and Hughes, 2016) was used as a reference to determine the copy number of *pmrD* (Hjort et al., 2016). The sequences of primers used in this study are listed in Table 4. The assay was performed in 20 μL reaction volumes with 10 μL of Fast SYBR™ Green Master Mix (Thermo Fisher Scientific), 0.5 μL each of the forward and reverse primer at 10 μM concentration, and 2 μL of DNA template in one of following dilutions: undiluted, 1:10 and 1:100 in RT-PCR grade water. Threshold cycle (C_T) values for the two genes at each dilution (serving as triplicates) were measured using quantitative PCR (qPCR) on a StepOnePlus Real-Time PCR System (Applied Biosystems, Carlsbad, CA, USA). Thermal cycling conditions used have been previously described (Yang et al., 2014). The gene copy number was determined by the formula: *pmrD* gene copy number $= 2^{CT(hcaT)\ -\ CT(pmrD)}$ (Hjort et al., 2016).

2.7. Statistical analysis

Unpaired *t*-test with a significance level of $P < 0.05$ was performed using the web-based GraphPad Software (https://www.graphpad.com/) to compare the OD_{595} increment values at each concentration of colistin and the *pmrD* gene copy numbers. Each data point of the OD_{595} increment values was derived from a minimum of duplicates with three independent trials. The gene copy numbers were determined from two biological replicates with triplicate measurements each.

3. Results

3.1. AMR phenotype-genotype disagreements

Table 1 presents the MICs and AST profiles of the strains used in the study while Table 2 presents the list of all genetic resistance determinants in each bacterial strain obtained from ResFinder 3.2.0 analysis together with the corresponding predicted AMR phenotypes according to literature. Comparing the resistance phenotypes in Table 1 to the genotypically predicted phenotypes in Table 2, all instances of AMR phenotype-genotype disagreements were obtained, compiled and presented in Table 3.

A total of five instances of false negative errors were reported. The B27 and WT5 strains were found to be carrying the β–lactamase genes bla_{TEM-1B} and bla_{CARB-2}, respectively (Table 2). The bla_{TEM-1B} and bla_{CARB-2} genes belong to the molecular class A of the β–lactamase family and are completely inhibited by clavulanate β–lactamase inhibitor (Bush et al., 1995). As a result, the anomalous Aug resistance in these two strains were instances of false negative errors (Table 3), i.e. phenotypic resistance without underlying genetic resistance determinant responsible. B27 did not have any genetic resistance determinant for Cl despite its phenotypic resistance (Tables 1 and 2) and is an instance of false negative error (Table 3). Other instances of false negative errors were observed in WD17 which was Am-resistant and in WW28B which was Fos-resistant but both without their respective resistance genes (Tables 1 and 2).

Apart from false negative errors, the alternative scenarios where phenotypic resistance was absent despite the presence of the genetic resistance determinant, i.e. false positives, were also observed. Particularly, the *aac(6')-Iaa* gene responsible for amikacin (Ak) and tobramycin (To) resistance was widely observed in B27, M24, WT5 and

Table 2
List of resistance determinants and their corresponding predicted resistance phenotypes of strains used in this study. Bold highlight indicates genes that were referred to and discussed in the text.

Strain ID	Resistance Determinants	Predicted Resistance Phenotype
B27	*aac(3)-IV*	Gm, To
	aac(6')-Iaa	Ak, To
	aadA1	S[a]
	aph(4)-Ia	Hyg[a]
	bla_{TEM-1B}	Am, Pi
	dfrA15	T/S
	floR	C
	fosA4	Fos
	mph(A)	Azm[a]
	qnrS1	Reduced susceptibility to fluoroquinolone
	sul3	T/S
	tet(A)	Te
B28	***aac(6')-Iaa***	Ak, To
M24	***aac(6')-Iaa***	Ak, To
	aph(3')-Ia	K[a]
	$bla_{TEM-176}$	Am, Pi
	dfrA14	T/S
	floR	C
	mcr-1	Cl
	parC T57S	Nal[a]; Reduced susceptibility to fluoroquinolones
	qnrS1	Reduced susceptibility to fluoroquinolones
	tet(A)	Te
WD17	***aac(6')-Iaa***	Ak, To
WT5	***aac(6')-Iaa***	Ak, To
	bla_{CARB-2}	Am, Pi
	dfrA1	T/S
	floR	C
	gyrA D87N	Nal[a]; Reduced susceptibility to fluoroquinolones
	parC T57S	Nal[a]; Reduced susceptibility to fluoroquinolones
	sul1	T/S
	tet(G)	Te
WW28B	***aac(6')-Iaa***	Ak, To
	aadA1, aadA2	S[a]
	aph(3'')-Ib	S[a]
	aph(6)-Id	S[a]
	bla_{TEM-1B}	Am, Pi
	catA2, floR	C
	dfrA12	T/S
	mph(A)	Azm[a]
	parC T57S	Nal[a]; Reduced susceptibility to fluoroquinolones
	qnrS1	Reduced susceptibility to fluoroquinolones
	sul1, sul3	T/S
	tet(A), tet(M)	Te

Abbreviations: Ak, amikacin; Am, ampicillin; Aug, amoxicillin/clavulanate; C, chloramphenicol; Azm, aztreonam; Cl, colistin; Fos, fosfomycin; Gm, gentamicin; Hyg, hygromycin B; K, kanamycin; Nal, nalidixic acid; S, streptomycin; Te, tetracycline; To, tobramycin; T/S, trimethoprim/sulfamethoxazole.
[a] Represents antimicrobials not tested in this study.

Table 1
Strain ID, serotype, and select MIC and phenotype interpretations of *S. enterica* isolates used in this study. Antimicrobials that are grouped in the same box belong to the same class.

Strain ID	Serotype	MIC (μg/mL) and Phenotype											
		Am	Pi	Aug	P/T	C	Cl	Fos	T/S	Ak	Gm	To	Te
B27	Saintpaul	> 8 (R)	> 16 (R)	> 8/4 (R)	≤4 (S)	> 8 (R)	> 2 (R)	> 32 (R)	> 4/76 (R)	≤8 (S)	> 4 (R)	> 4 (R)	> 8 (R)
B28	Saintpaul	1 (S)	≤4 (S)	≤2/1 (S)	≤4 (S)	≤8 (S)	≤2 (S)	≤32 (S)	≤2/38 (S)	≤8 (S)	≤2 (S)	≤2 (S)	≤4 (S)
M24	Brancaster	> 8 (R)	> 16 (R)	4/2 (S)	≤4 (S)	> 8 (R)	> 2 (R)	≤32 (S)	> 4/76 (R)	≤8 (S)	≤2 (S)	≤2 (S)	> 8 (R)
WD17	Gaminara	> 8 (R)	16 (I)	≤2/1 (S)	≤4 (S)	≤8 (S)	≤2 (S)	≤32 (S)	≤2/38 (S)	≤8 (S)	≤2 (S)	≤2 (S)	≤4 (S)
WT5	Albany	> 8 (R)	> 16 (R)	> 8/4 (R)	≤4 (S)	> 8 (R)	≤2 (S)	≤32 (S)	> 4/76 (R)	≤8 (S)	≤2 (S)	≤2 (S)	> 8 (R)
WW28B	Stanley	> 8 (R)	> 16 (R)	8/4 (S)	≤4 (S)	> 8 (R)	≤2 (S)	> 32 (R)	> 4/76 (R)	≤8 (S)	≤2 (S)	≤2 (S)	> 8 (R)

Abbreviations: Ak, amikacin; Am, ampicillin; Aug, amoxicillin/clavulanate; C, chloramphenicol; Cl, colistin; Fos, fosfomycin; Gm, gentamicin; Pi, piperacillin; P/T, piperacillin/tazobactam; Te, tetracycline; To, tobramycin; T/S, trimethoprim/sulfamethoxazole.

Table 3
List of genotype-phenotype disagreements detected in this study and the nature of errors.

Strain ID	Nature of Error	Genotype-Phenotype Disagreement
B27	False negative	Aug^R without any resistance determinant
	False negative	Cl^R without any resistance determinant
	False positive	Ak^S despite presence of *aac(6′)-Iaa* gene
B28	False positive	Ak^S, To^S despite presence of *aac(6′)-Iaa* gene
M24	False positive	Ak^S, To^S despite presence of *aac(6′)-Iaa* gene
WD17	False negative	Am^R without any resistance determinant
	False positive	Ak^S, To^S despite presence of *aac(6′)-Iaa* gene
WT5	False negative	Aug^R without any resistance determinant
	False positive	Ak^S, To^S despite presence of *aac(6′)-Iaa* gene
WW28B	False negative	Fos^R without any resistance determinant
	False positive	Ak^S, To^S despite presence of *aac(6′)-Iaa* gene

Abbreviation: Ak, amikacin; Am, ampicillin; Aug, amoxicillin/clavulanate; Cl, colistin; Fos, fosfomycin; To, tobramycin.
^{S}Represents susceptibility.
^{R}Represents resistance.

WW28B without the corresponding phenotypic resistance (Tables 1 and 2). The false positive errors were out of the scope of this study and were subsequently not discussed.

3.2. Heteroresistance and corresponding false negative WGS prediction errors

For every one of the five instances of false negative errors described in Section 3.1, a corresponding instance of heteroresistance was found. B27 and WT5 were heteroresistant to Aug as shown by the growth of colonies in the clear zone of inhibition (Fig. 1A and B, respectively). Likewise, WD17 was heteroresistant to Am (Fig. 1C) while WW28B was to Fos (Fig. 1D).

Heteroresistance toward Cl in B27 was tested using the PAP assay and the results are presented in Fig. 2. The *mcr-1* carrier M24 strain showed a relatively constant growth over 0.125–1 μg/mL due to its homogenous response toward Cl with an MNIC value of 1 μg/mL and an MIC of 4 μg/mL (resistance breakpoint of Cl). This indicates a 4-fold difference between MNIC and MIC. The Cl-susceptible B28 strain had an MNIC and MIC of 1 and 2 μg/mL with a 2-fold difference between the two. The growth of the B27 strain was observed to gradually decrease over 0.25–2 μg/mL, indicating a heterogeneous response toward Cl. With an MNIC and MIC of 0.25 and 4 μg/mL, respectively, the B27 strain had a 16-fold difference between the values. Heteroresistance was considered when the MIC is at least 8-fold or higher than the MNIC as previously suggested (El-Halfawy and Valvano, 2013), thereby, indicating Cl heteroresistance of B27.

3.3. Case study on pmrD gene copy number and colistin heteroresistance

To further explore the underlying reason for the failure of WGS to detect resistance due to the heteroresistance phenomenon, the molecular mechanism of Cl heteroresistance was chosen as a further case study. The underlying genetic mechanism of Cl-heteroresistance has previously been attributed in *S.* Typhimurium to the subpopulation which possesses the ability to temporarily, chromosomally amplify the copy numbers of *pmrD* gene which produces a positive regulator to a lipid A-modifying gene in response to increasing antimicrobial stress from Cl (Hjort et al., 2016).

The cell population of B28, M24, and B27, when grown in TSB without Cl, was found to be carrying an average *pmrD* copy number of 1.7, 1.0 and 1.3, respectively (Fig. 3). Upon exposure to Cl, the average *pmrD* copy numbers in B27 were significantly amplified to 4.3 copies while remaining at a constant 1.0 copy in M24 (Fig. 3). This suggests that in response to sublethal Cl exposure, B27 chromosomally amplified the copy number of *pmrD* above 4 copies, the cut-off for conferring heteroresistance in *S.* Typhimurium previously described by Hjort et al. (2016) at which the downstream lipid A-modification became significant to confer Cl resistance.

4. Discussion

In the clinical arena, heteroresistance has long been known to negatively impact the clinical outcome of antimicrobial therapy (Campanile et al., 2012; Hernan et al., 2009; Sola et al., 2011), resulting in therapeutic failure. This may arise when the administered dose of the antimicrobial is sufficient to only eliminate the majority of the sensitive population, leaving behind the highly resistant sub-population to proliferate (El-Halfawy and Valvano, 2015). However, these findings highlight that heteroresistance may also present significant challenges in the area of surveillance especially now that the use of WGS to monitor AMR across a large number of isolates is becoming more commonplace.

This study presented five instances of heteroresistance in *S. enterica* in conjunction with the corresponding failure of WGS to predict the phenotypic resistance in these instances due to a lack of genotypic resistance determinants. To better understand the underlying reason for heteroresistance evading detection by WGS, a further case study of molecular mechanisms of Cl heteroresistance involving the modulation of *pmrD* gene copy number was performed.

Failure of WGS to detect resistance in the heteroresistant subpopulation of B27 can be due to several reasons. Firstly, since the prediction of AMR by WGS seeks to correlate a permanent genetic feature (such as an acquired AMR gene or a specific mutation) to a particular resistance, resistance mediated by unstable genetic features such as temporary amplification of *pmrD* copy number could not be reliably employed by WGS to predict phenotype. Secondly, the repetitive nature of gene copy amplification presents technical challenges in sequence alignment and assembly, especially in next-generation sequencing (NGS) platforms utilized in this study due to its short length of the reads generated (Treangen and Salzberg, 2012). In addition to creating gaps in the genome during assembly (Treangen and Salzberg, 2012), the reads from the multiple copies may also be erroneously collapsed on top of one another (Phillippy et al., 2008; Pop and Salzberg, 2008), thereby, obscuring the true copy number of the gene in the final assembled genome. Lastly, even if the advancements in bioinformatics software and sequencing technology allow us to reliably and accurately determine, in this case, the *pmrD* copy number in the genome of B27 strain, it necessitates the harvest of the B27 cells "in the act of" expressing that temporary genetic feature in their genome. This means that B27 must first be grown in the presence of Cl to induce the *pmrD* amplification before being sequenced. Therefore, by definition, the knowledge of their heteroresistant status toward Cl must precede

Table 4
List of primers used in this study.

Gene	Function	Sequence (5′ to 3′)	Amplicon Size (bp)
hcaT	Single copy reference gene	F: GCCGTGGCTGATTGTGATA R: CCTGCAAACGAATCACCT	113
pmrD	Positive regulator to lipid A modifying gene	F: GCGATATCCTGTCGCCTTTAC R: CCATTCGTCGCAGGAATAAC	104

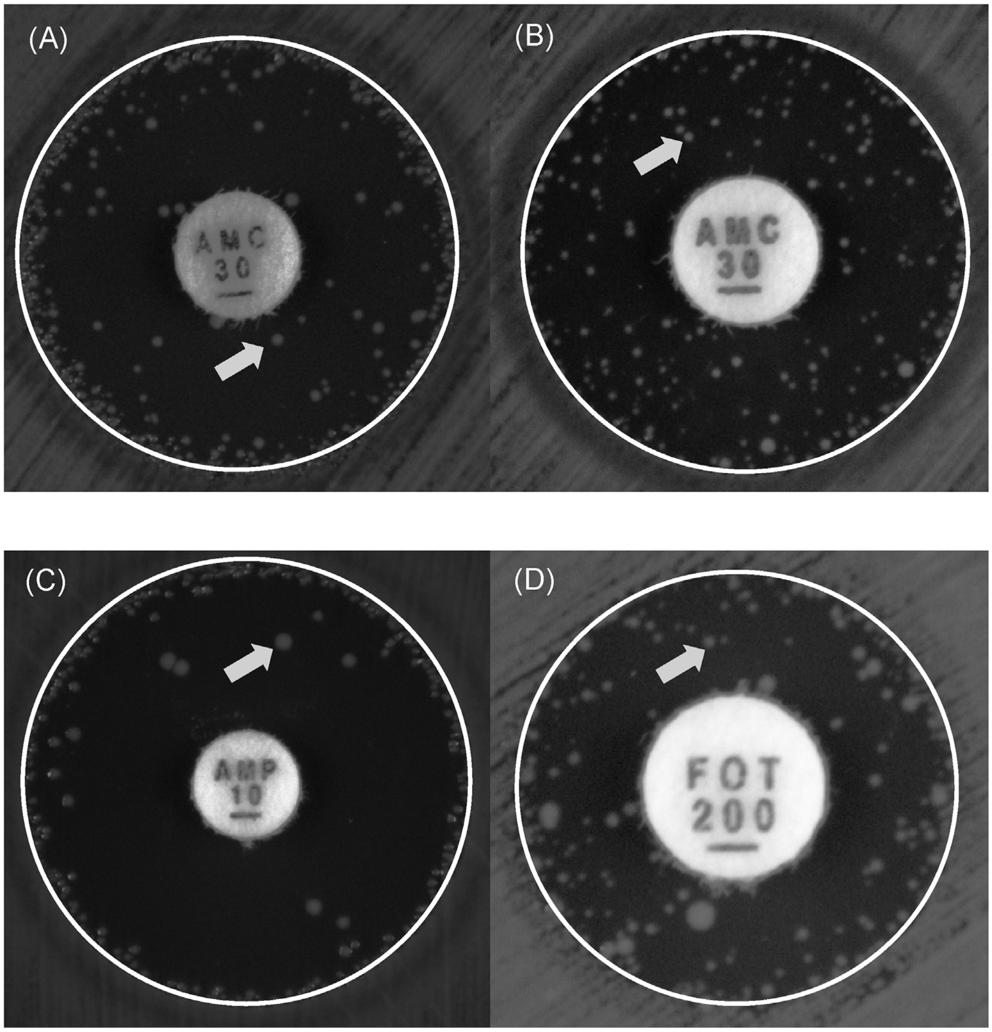

Fig. 1. Heteroresistance, as demonstrated by the growth of colonies in the clear zone by disk diffusion test in (A) B27 (B) WT5 for amoxicillin/clavulanate (Aug) (C) WD17 for ampicillin (Am), (D) WW28B for fosfomycin (Fos). Arrows highlight examples of heteroresistant isolates.

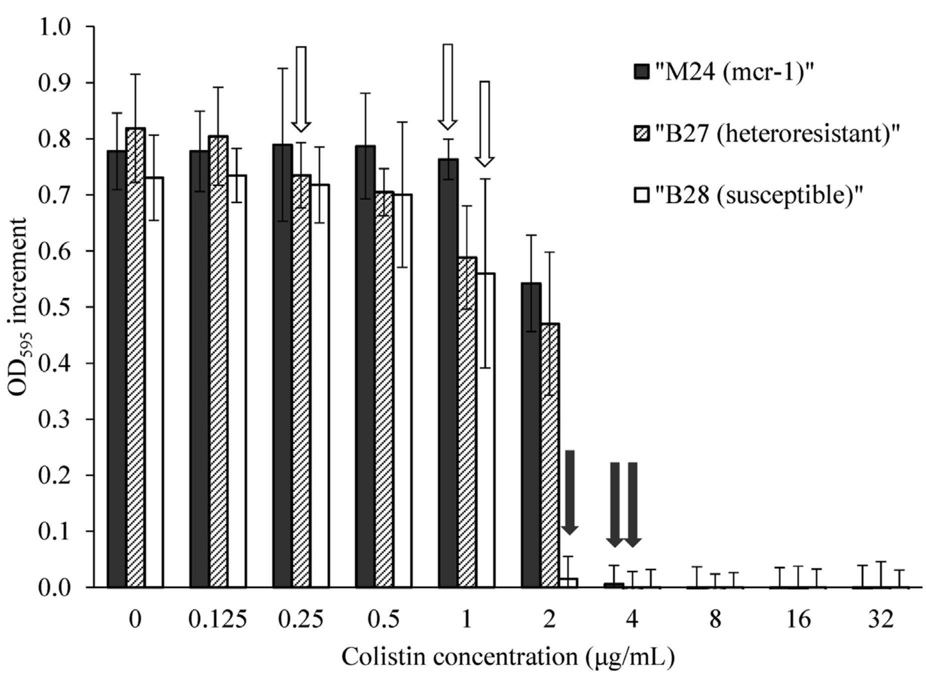

Fig. 2. Population analysis profiling (PAP) assay of M24 (*mcr-1* carrier), B27 (colistin heteroresistant), and B28 (colistin susceptible control). Open arrows represent maximum non-inhibitory concentration (MNIC). Solid arrows represent minimum inhibitory concentration (MIC). MNIC and MIC fold difference of ≥8-fold was considered as evidence of heteroresistance.

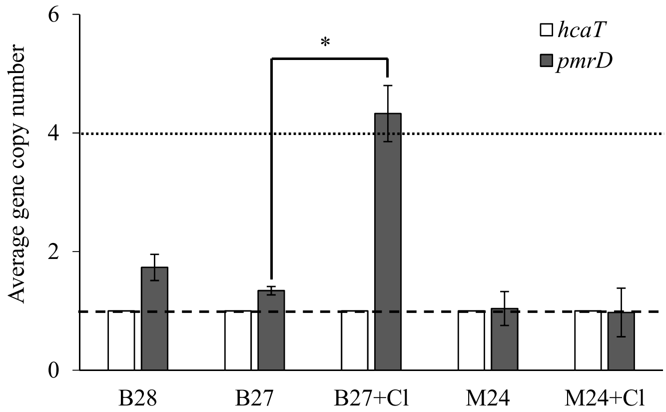

Fig. 3. Average gene copy number (± SD) of various isolates when grown in the absence and in presence of 2 μg/mL colistin (1/2 MIC), latter denoted by "+Cl". The dashed line represents 1 gene copy number of the chromosomal *hcaT* gene used as reference. The dotted line represents the *pmrD* gene copy number (~4 copies) at and above which heteroresistance to colistin has previously been demonstrated (22). * indicates the significant difference ($P < 0.01$).

the detection of the underlying temporary genetic feature, rendering WGS to have little value as a predictive tool in this particular scenario.

A recent study reported that a high prevalence of heteroresistance in pathogenic *Escherichia coli*, *Klebsiella pneumoniae*, *S.* Typhimurium, *Acinetobacter baumannii* were mainly caused by unstable, spontaneous tandem amplification of various genes within subpopulations which ultimately resulted in heteroresistance (Nicoloff et al., 2019). Genetic changes of this nature are precisely the type that WGS fail to detect due to reasons already discussed above in this study; they are temporary, involve tandem repeats, and are observed only upon exposure to antibiotics. Since these changes are now known to constitute a majority of heteroresistant instances (Nicoloff et al., 2019), it can be concluded that WGS may, therefore, fail to detect a majority of heteroresistant instances in *S. enterica* and possibly in other bacteria resulting in false negative predictions.

These findings may disproportionately affect the surveillance of AMR in food and environmental isolates of *S. enterica* than the diagnostics of resistance in clinical samples for treatment purposes. Challenges that are preventing WGS from becoming the primary diagnostic tool to diagnose AMR in clinical isolates are out of the scope of this study and has been discussed elsewhere (Köser et al., 2012). In the surveillance of AMR in food and environmental isolates, however, often the sheer number of isolates may make phenotypic testing on a large number of antimicrobials impractical. Hence, WGS is increasingly being relied upon to monitor the AMR patterns in these large number of isolates. The AMR patterns yielded from large retrospective studies using WGS involving thousands of food and environmental isolates collected over time and different geographical locations may shed light on the historical evolution of the AMR and offer insights into the emergence of AMR in the future. However, a caveat is that with increasing reliance on WGS, the impact of heteroresistance in causing false negative errors described in this study will become increasingly apparent. This may lead to erroneous interpretations of data in the surveillance of these food and environmental isolates.

Cl resistance is of clinical significance as polymyxins such as Cl is the last line of defence against many multidrug-resistant Gram-negative bacilli (Paterson and Harris, 2016). First-ever instances of Cl-resistant isolates have been reported in food animals and raw meat in China (Liu et al., 2016), Denmark (Hasman et al., 2015), Vietnam (Malhotra-Kumar et al., 2016), Brazil (Monte et al., 2017), to name a few, and with this study, in Singapore from *S. enterica* isolated from raw chicken.

Although the original intention of replicating the determination of the *pmrD* copy number in B27 was to provide the genetic evidence of heteroresistance to be presented in conjunction with the false negative WGS prediction, a number of new insights with regards to Cl heteroresistance in *S. enterica* were observed. Firstly, the aforementioned genetic mechanism has, to our knowledge, so far only been described in *S.* Typhimurium (Hjort et al., 2016; Nicoloff et al., 2019) among *S. enterica*, while the present study described the identical genetic mechanism in *S.* Saintpaul, suggesting that this mechanism might be more ubiquitous across the different serotypes of *S. enterica* than previously known. Secondly, in the present study, it was demonstrated that one passage at 2 μg/mL of Cl was sufficient to amplify the average *pmrD* copy numbers in B27 from 1.3 to 4.3 copies and that the latter copy number was sufficient to phenotypically raise the MIC of the overall population to 4 μg/mL, thereby, effectively rendering B27 resistant to Cl despite the lack of acquired Cl-resistance genes such as *mcr-1*. In comparison, the previous study reported that the median copy number of 6 copies in *S.* Typhimurium was achieved after serial passages at 1 μg/mL of Cl for 20 generations (Hjort et al., 2016). Findings from this study suggest that Cl-heteroresistant *S. enterica* not only can amplify *pmrD* to increase resistance to Cl but might also be able to do so in a rapid manner in response to Cl exposure within just one generation. However, whether this rapid response is ubiquitous across every heteroresistant isolate is unclear.

5. Conclusion

In light of the empirical evidence in the present study linking heteroresistance to false negative errors in predicting AMR from WGS data, and the reasons discussed, it can be concluded that WGS failed to detect instances of resistance due to heteroresistant subpopulations in *S. enterica*. Thus, the heteroresistance phenomenon poses a serious limitation on using WGS as a sole routine AMR determination tool as doing so might severely underestimate the true phenotypic resistance rate caused by heteroresistance. This may disproportionately impact AMR surveillance of food and environmental *S. enterica* isolates where WGS may more heavily be relied upon to determine AMR patterns.

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgement

The authors would like to thank Dr Pimlapas Leekitcharoenphon and Professor Rene Hendriksen from the National Food Institute, the Technical University of Denmark for guidance on the use of bioinformatics tools. This work was jointly supported by the National University of Singapore, Singapore Centre for Environmental Life Sciences Engineering, Nanyang Technological University using the AcRF Tier 1 (MOE2015-T1-002-100) grant from Ministry of Education, Singapore and the National Environment Agency, Singapore in 2017 and also by the Korea National University of Transportation in 2018.

References

Bankevich, A., Nurk, S., Antipov, D., Gurevich, A.A., Dvorkin, M., Kulikov, A.S., Lesin, V.M., Nikolenko, S.I., Pham, S., Prjibelski, A.D., 2012. SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. J. Comput. Biol. 19, 455–477.

Brandis, G., Hughes, D., 2016. The selective advantage of synonymous codon usage bias in *Salmonella*. PLoS Genet. 12, e1005926.

Bush, K., Jacoby, G.A., Medeiros, A.A., 1995. A functional classification scheme for beta-lactamases and its correlation with molecular structure. Antimicrob. Agents Chemother. 39, 1211–1233.

Campanile, F., Bongiorno, D., Falcone, M., Vailati, F., Pasticci, M., Perez, M., Raglio, A., Rumpianesi, F., Scuderi, C., Suter, F., 2012. Changing Italian nosocomial-community trends and heteroresistance in *Staphylococcus aureus* from bacteremia and endocarditis. Eur. J. Clin. Microbiol. Infect. Dis. 31, 739–745.

Clausen, P.T., Aarestrup, F.M., Lund, O., 2018. Rapid and precise alignment of raw reads against redundant databases with KMA. BMC Bioinf. 19, 307.

Clausen, P.T., Zankari, E., Aarestrup, F.M., Lund, O., 2016. Benchmarking of methods for identification of antimicrobial resistance genes in bacterial whole genome data. J. Antimicrob. Chemother. 71, 2484–2488.

CLSI, 2017. Performance Standards for Antimicrobial Susceptibility Testing M100-S27, Zone Diameter and Minimal Inhibitory Concentration Breakpoints for Enterobacteriaceae. Clinical and Laboratory Standards Institute, pp. 32–41.

Didelot, X., Bowden, R., Wilson, D.J., Peto, T.E., Crook, D.W., 2012. Transforming clinical microbiology with bacterial genome sequencing. Nat. Rev. Genet. 13, 601–612.

El-Halfawy, O.M., Valvano, M.A., 2013. Chemical communication of antibiotic resistance by a highly resistant subpopulation of bacterial cells. PloS One 8, e68874.

El-Halfawy, O.M., Valvano, M.A., 2015. Antimicrobial heteroresistance: an emerging field in need of clarity. Clin. Microbiol. Rev. 28, 191–207.

EUCAST, 2018. Breakpoint Tables for Interpretation of MICs and Zone Diameters. Version 8. 0 European Committee on Antimicrobial Susceptibility Testing.

Gordon, N., Price, J., Cole, K., Everitt, R., Morgan, M., Finney, J., Kearns, A., Pichon, B., Young, B., Wilson, D., 2014. Prediction of *Staphylococcus aureus* antimicrobial resistance from whole genome sequencing. J. Clin. Microbiol. 52, 1182–1191.

Hasman, H., Hammerum, A.M., Hansen, F., Hendriksen, R.S., Olesen, B., Agersø, Y., Zankari, E., Leekitcharoenphon, P., Stegger, M., Kaas, R.S., 2015. Detection of *mcr-1* encoding plasmid-mediated colistin-resistant *Escherichia coli* isolates from human bloodstream infection and imported chicken meat, Denmark 2015. Eurosurveillance 20, 30085.

Hernan, R.C., Karina, B., Gabriela, G., Marcela, N., Carlos, V., Angela, F., 2009. Selection of colistin-resistant *Acinetobacter baumannii* isolates in postneurosurgical meningitis in an intensive care unit with high presence of heteroresistance to colistin. Diagn. Microbiol. Infect. Dis. 65, 188–191.

Hjort, K., Nicoloff, H., Andersson, D.I., 2016. Unstable tandem gene amplification generates heteroresistance (variation in resistance within a population) to colistin in *Salmonella enterica*. Mol. Microbiol. 102, 274–289.

Jo, A., Ding, T., Ahn, J., 2017. Comparison of antibiotic resistance phenotypes in laboratory strains and clinical isolates of *Staphylococcus aureus, Salmonella* Typhimurium, and *Klebsiella pneumoniae*. Food Sci. Biotechnol. 26, 1773–1779.

Joensen, K.G., Tetzschner, A.M., Iguchi, A., Aarestrup, F.M., Scheutz, F., 2015. Rapid and easy i*n silico* serotyping of *Escherichia coli* using whole genome sequencing (WGS) data. J. Clin. Microbiol. 53, 2410–2426.

Kahlmeter, G., 2014. Defining antibiotic resistance-towards international harmonization. Ups. J. Med. Sci. 119, 78–86.

Köser, C.U., Ellington, M.J., Cartwright, E.J., Gillespie, S.H., Brown, N.M., Farrington, M., Holden, M.T., Dougan, G., Bentley, S.D., Parkhill, J., 2012. Routine use of microbial whole genome sequencing in diagnostic and public health microbiology. PLoS Pathog. 8.

Larsen, M.V., Cosentino, S., Rasmussen, S., Friis, C., Hasman, H., Marvig, R.L., Jelsbak, L., Pontén, T.S., Ussery, D.W., Aarestrup, F.M., 2012. Multilocus sequence typing of total genome sequenced bacteria. J. Clin. Microbiol. 50, 1355–1361.

Leekitcharoenphon, P., Nielsen, E.M., Kaas, R.S., Lund, O., Aarestrup, F.M., 2014. Evaluation of whole genome sequencing for outbreak detection of *Salmonella enterica*. PloS One 9, e87991.

Liu, Y.Y., Wang, Y., Walsh, T.R., Yi, L.X., Zhang, R., Spencer, J., Doi, Y., Tian, G., Dong, B., Huang, X., 2016. Emergence of plasmid-mediated colistin resistance mechanism *mcr-1* in animals and human beings in China: a microbiological and molecular biological study. Lancet Infect. Dis. 16, 161–168.

Lo-Ten-Foe, J.R., de Smet, A.M.G., Diederen, B.M., Kluytmans, J.A., van Keulen, P.H., 2007. Comparative evaluation of the VITEK 2, disk diffusion, Etest, broth microdilution, and agar dilution susceptibility testing methods for colistin in clinical isolates, including heteroresistant *Enterobacter cloacae* and *Acinetobacter baumannii* strains. Antimicrob. Agents Chemother. 51, 3726–3730.

Malhotra-Kumar, S., Xavier, B.B., Das, A.J., Lammens, C., Hoang, H.T.T., Pham, N.T., Goossens, H., 2016. Colistin-resistant *Escherichia coli* harbouring *mcr-1* isolated from food animals in Hanoi, Vietnam. Lancet Infect. Dis. 16, 286–287.

McDermott, P.F., Tyson, G.H., Kabera, C., Chen, Y., Li, C., Folster, J.P., Ayers, S.L., Lam, C., Tate, H.P., Zhao, S., 2016. The use of whole genome sequencing for detecting antimicrobial resistance in nontyphoidal *Salmonella*. Antimicrob. Agents Chemother. 60, 5515–5520.

Monte, D.F., Mem, A., Fernandes, M.R., Cerdeira, L., Esposito, F., Galvão, J.A., Franco, B.D., Lincopan, N., Landgraf, M., 2017. Chicken meat as reservoir of colistin-resistant *Escherichia coli* carrying *mcr-1* genes in South America. Antimicrob. Agents Chemother. 61 e02718-2716.

Neuert, S., Nair, S., Day, M.R., Doumith, M., Ashton, P.M., Mellor, K.C., Jenkins, C., Hopkins, K.L., Woodford, N., de Pinna, E., 2018. Prediction of phenotypic antimicrobial resistance profiles from whole genome sequences of non-typhoidal *Salmonella enterica*. Front. Microbiol. 9, 592.

Nicoloff, H., Hjort, K., Levin, B.R., Andersson, D.I., 2019. The high prevalence of antibiotic heteroresistance in pathogenic bacteria is mainly caused by gene amplification. Nat. Microbiol. 4, 504–514.

Paterson, D.L., Harris, P.N., 2016. Colistin resistance: a major breach in our last line of defence. Lancet Infect. Dis. 16, 132–133.

Phillippy, A.M., Schatz, M.C., Pop, M., 2008. Genome assembly forensics: finding the elusive mis-assembly. Genome Biol. 9, R55.

Pop, M., Salzberg, S.L., 2008. Bioinformatics challenges of new sequencing technology. Trends Genet. 24, 142–149.

Pournaras, S., Kristo, I., Vrioni, G., Ikonomidis, A., Poulou, A., Petropoulou, D., Tsakris, A., 2010. Characteristics of meropenem heteroresistance in *Klebsiella pneumoniae* carbapenemase (KPC)-producing clinical isolates of *K. pneumoniae*. J. Clin. Microbiol. 48, 2601–2604.

Punina, N., Makridakis, N., Remnev, M., Topunov, A., 2015. Whole-genome sequencing targets drug-resistant bacterial infections. Hum. Genom. 9, 19.

Reller, L.B., Weinstein, M., Jorgensen, J.H., Ferraro, M.J., 2009. Antimicrobial susceptibility testing: a review of general principles and contemporary practices. Clin. Infect. Dis. 49, 1749–1755.

Shaw, K., Rather, P., Hare, R., Miller, G., 1993. Molecular genetics of aminoglycoside resistance genes and familial relationships of the aminoglycoside-modifying enzymes. Microbiol. Rev. 57, 138–163.

Sola, C., Lamberghini, R.O., Ciarlantini, M., Egea, A.L., Gonzalez, P., Diaz, E.G., Huerta, V., Gonzalez, J., Corso, A., Vilaro, M., 2011. Heterogeneous vancomycin-intermediate susceptibility in a community-associated methicillin-resistant *Staphylococcus aureus* epidemic clone, in a case of infective endocarditis in Argentina. Ann. Clin. Microbiol. Antimicrob. 10, 15.

Stoesser, N., Batty, E., Eyre, D., Morgan, M., Wyllie, D., Del Ojo Elias, C., Johnson, J., Walker, A., Peto, T., Crook, D., 2013. Predicting antimicrobial susceptibilities for *Escherichia coli* and *Klebsiella pneumoniae* isolates using whole genomic sequence data. J. Antimicrob. Chemother. 68, 2234–2244.

Treangen, T.J., Salzberg, S.L., 2012. Repetitive DNA and next-generation sequencing: computational challenges and solutions. Nat. Rev. Genet. 13, 36–46.

Tyson, G.H., McDermott, P.F., Li, C., Chen, Y., Tadesse, D.A., Mukherjee, S., Bodeis-Jones, S., Kabera, C., Gaines, S.A., Loneragan, G.H., 2015. WGS accurately predicts antimicrobial resistance in *Escherichia coli*. J. Antimicrob. Chemother. 70, 2763–2769.

Uddin, M.J., Anh, J., 2018. Characterization of β-lactamase- and efflux pump-mediated multiple antibiotic resistance in *Salmonella* yphimurium. Food Sci. Biotechnol. 27, 921–928.

Yang, Y., Khoo, W.J., Zheng, Q., Chung, H.J., Yuk, H.G., 2014. Growth temperature alters *Salmonella* Enteritidis heat/acid resistance, membrane lipid composition and stress/virulence related gene expression. Int. J. Food Microbiol. 172, 102–109.

Zankari, E., Hasman, H., Cosentino, S., Vestergaard, M., Rasmussen, S., Lund, O., Aarestrup, F.M., Larsen, M.V., 2012. Identification of acquired antimicrobial resistance genes. J. Antimicrob. Chemother. 67, 2640–2644.

Zhang, S., Yin, Y., Jones, M.B., Zhang, Z., Kaiser, B.L.D., Dinsmore, B.A., Fitzgerald, C., Fields, P.I., Deng, X., 2015. *Salmonella* serotype determination utilizing high-throughput genome sequencing data. J. Clin. Microbiol. 53, 1685–1692.

Zwe, Y.H., Tang, V.C.Y., Aung, K.T., Gutiérrez, R.A., Ng, L.C., Yuk, H.G., 2018. Prevalence, sequence types, antibiotic resistance and, *gyrA* mutations of *Salmonella* isolated from retail fresh chicken meat in Singapore. Food Contr. 90, 233–240.